



*Commonwealth of Virginia*

*VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY*

[www.deq.virginia.gov](http://www.deq.virginia.gov)

David L. Bulova  
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus  
Director

March 26<sup>th</sup>, 2026

Mr. Robb Truedinger  
Authorized Representative  
Amazon Data Services  
13820 Sunrise Valley Drive  
Herndon, VA 20171

Location: Prince William County  
Registration No.: 74333

Dear Mr. Truedinger:

Attached is a permit to construct and operate engine generator-sets (gen-sets) at Amazon Data Services' IAD-45 data center in accordance with the provisions of the Commonwealth of Virginia State Air Pollution Control Board (Board's) Regulations for the Control and Abatement of Air Pollution (Regulations).

In the course of evaluating the application and arriving at a final decision to approve the stationary source, the Department of Environmental Quality (DEQ) deemed the application complete on March 25<sup>th</sup>, 2026.

This permit contains legally enforceable conditions. Failure to comply may result in a Notice of Violation and/or civil charges. Please read all permit conditions carefully.

This permit approval to construct and operate shall not relieve Amazon Data Services, Inc. of the responsibility to comply with all other local, state, and federal permit regulations.

The diesel engine gen-sets may be subject to 40 CFR 60, New Source Performance Standard (NSPS) Subpart IIII – *Standards of Performance for Stationary Compression Ignition Internal Combustion Engines* and 40 CFR 63, National Emission Standards for Hazardous Air Pollutants for Source Categories (NESHAP) Subpart ZZZZ – *National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*. In summary, the units may be required to comply with certain federal emission standards and operating limitations. DEQ advises you to review the referenced NESHAP and NSPS to ensure compliance with applicable emission and operational limitations. As the owner/operator, you are also responsible for any monitoring, notification, reporting and recordkeeping requirements of the NESHAP and NSPS. Notifications shall only be sent to EPA, Region III.

To review any federal rules referenced in the above paragraph or in the attached permit, the US Government Publishing Office maintains the text of these rules at [www.ecfr.gov](http://www.ecfr.gov), Title 40, Part 60 and 63.

The Board's Regulations as contained in Title 9 of the Virginia Administrative Code 5-170-200 provide that you may request a formal hearing from this case decision by filing a petition with the Board within thirty days after this case decision notice was mailed or delivered to you. 9VAC5-170-200 provides that you may request direct consideration of the decision by the Board if the Director of DEQ made the decision. Please consult the relevant regulations for additional requirements for such requests.

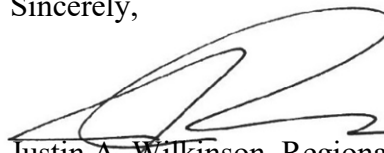
As provided by Rule 2A:2 of the Supreme Court of Virginia, you have thirty days from the date you actually received this permit or the date on which it was mailed to you, whichever occurred first, within which to initiate an appeal of this decision by filing a Notice of Appeal with:

Michael S. Rolband, Director  
Department of Environmental Quality  
P. O. Box 1105  
Richmond, VA 23218

If this permit was delivered to you by mail, three days are added to the thirty-day period in which to file an appeal. Please refer to Part Two A of the Rules of the Supreme Court of Virginia for information on the required content of the Notice of Appeal and for additional requirements governing appeals from decisions of administrative agencies.

If you have any questions concerning this permit, please contact Ms. Kaitlin King at (804) 338-2430 or [kaitlin.king@deq.virginia.gov](mailto:kaitlin.king@deq.virginia.gov).

Sincerely,



Justin A. Wilkinson, Regional Air Permit Manager  
Virginia Department of Environmental Quality  
[justin.wilkinson@deq.virginia.gov](mailto:justin.wilkinson@deq.virginia.gov)  
Northern Regional Office  
13901 Crown Court, Woodbridge, VA 22193  
(703) 583-3800

JAW/KMK/74333 mNSR (2026-03-26)

Attachment: Permit



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David L. Bulova  
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus  
Director

**STATIONARY SOURCE PERMIT TO CONSTRUCT AND OPERATE**

In compliance with the Federal Clean Air Act and the Commonwealth of Virginia Regulations for the Control and Abatement of Air Pollution,

Amazon Data Services IAD-45  
13820 Sunrise Valley Drive  
Herndon, VA 20171  
Registration No.: 74333

is authorized to construct and operate

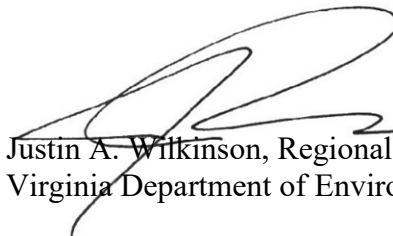
diesel engine generator sets (gen-sets)

located at

IAD-45  
8870 Hornbaker Rd  
Manassas, Virginia 20109  
(Prince William County)

in accordance with the Conditions of this permit.

Approved on March 26<sup>th</sup>, 2026.



Justin A. Wilkinson, Regional Air Permit Manager  
Virginia Department of Environmental Quality

Permit consists of 23 pages.

Permit Conditions 1 to 36

Attachment: Source Testing Report Format

## INTRODUCTION

This permit document is based on the following permit approval and the respective permit application:

- mNSR permit dated March 26<sup>th</sup>, 2026 based on the permit application dated September 8<sup>th</sup>, 2025 including supplemental information dated March 9<sup>th</sup>, 2026, March 18<sup>th</sup>, 2026 and March 25<sup>th</sup>, 2026.

Any changes in the permit application specifications or any existing facilities which alter the impact of the facility on air quality may require a permit. Failure to obtain such a permit prior to construction may result in enforcement action. In addition, this facility may be subject to additional applicable requirements not listed in this permit.

Words or terms used in this permit shall have meanings as provided in 9VAC5-80-1110 and 9VAC5-10-10 of the Commonwealth of Virginia State Air Pollution Control Board (Board's) Regulations (Regulations) for the Control and Abatement of Air Pollution. The regulatory reference or authority for each condition is listed in parentheses () after each condition. The enabling permit program, or permit programs is provided below each permit condition in the regulatory authority parenthetical as follows: 9VAC5-80-850 for Article 5, 9VAC5-80-1180 for Article 6, 9VAC5-80-1985 for Article 8, and 9VAC5-80-2050 for Article 9. The most recent effective date for a term or condition is listed in brackets [ ]. When identical conditions for one or more emission units are combined, the effective date listed in this permit does not alter the prior effective date(s) for any such conditions as issued in a previous permit action. In accordance with 9VAC5-80-1120F, any condition not marked as state-only enforceable (SOE) is state and federally enforceable.

Annual requirements to fulfill legal obligations to maintain current stationary source emissions data will necessitate a prompt response by the permittee to requests by the DEQ or the Board for information to include, as appropriate: process and production data; changes in control equipment; and operating schedules. Such requests for information from the DEQ will either be in writing or by personal contact.

The availability of information submitted to the DEQ or the Board will be governed by applicable provisions of the Freedom of Information Act, §§ 2.2-3700 through 2.2-3714 of the Code of Virginia, § 10.1-1314 (addressing information provided to the Board) of the Code of Virginia, and 9VAC5-170-60 of the State Air Pollution Control Board Regulations. Information provided to federal officials is subject to appropriate federal law and regulations governing confidentiality of such information.

**Equipment List** - Equipment at this facility consists of:

**Equipment to be Constructed:**

| Ref. No(s).  | Equipment Description                                                    | Maximum Rated Capacity (each) | Add-on Control Technology                                                           | Delegated Federal Requirements | Original Permit Date |
|--------------|--------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------|--------------------------------|----------------------|
| 1 through 22 | (22) Non-emergency diesel-fired engine gen-sets, Caterpillar Model 3516E | 2,750 ekW, 4,043 bhp          | Selective Catalytic Reduction (SCR) and Catalyzed Diesel Particulate Filter (cDPF)* | N/A                            | 3/26/2026            |
| 23           | (1) Emergency diesel-fired engine gen-set, Caterpillar Model C18         | 750 ekW, 1,112 bhp            | None                                                                                | N/A                            | 3/26/2026            |
| T1 and T2    | (2) Emergency diesel-fired engine gen-sets, Caterpillar Model 3516C      | 2,000 ekW, 2,937 bhp          | None                                                                                | N/A                            | 3/26/2026            |

\* Safety Power, ecoCube (9550-H3C29) SCR+cDPF or Miratech (SP-KIT-M3-SW550-TLI-23120476) SCR + cDPF.

Specifications included in the above table are for informational purposes only and do not form enforceable terms or conditions of the permit.

## PROCESS REQUIREMENTS

1. **Emission Controls** – Emissions from the emergency diesel engine-gen-sets (Ref. Nos. 23, T1 and T2) shall be controlled by the following:
  - a. Nitrogen oxides (NO<sub>x</sub>) emissions from each emergency diesel engine gen-set shall be controlled by engine design.
  - b. Carbon monoxide (CO), particulate matter (PM/PM<sub>10</sub>/PM<sub>2.5</sub>), volatile organic compounds (VOCs) and visible emissions from the emergency diesel engine gen-sets shall be controlled by the use of good operating practices and performing maintenance in accordance with the manufacturer recommendations. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not degrade the air emissions from the engines.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

2. **Emission Controls (SCR)** – Nitrogen oxide (NO<sub>x</sub>) emissions from the non-emergency engine gen-sets (Ref. Nos. 1 through 22) shall be controlled by closed loop Selective Catalytic Reduction (SCR). Each SCR system shall be equipped with a temperature probe to continuously monitor the exhaust temperature at the outlet of the entire SCR+cDPF emissions control system while the engine gen-set is operational. Engine exhaust gas shall be treated with urea when the catalyst bed exhaust temperature of 572°F is achieved, except for periods of start-up, shutdown, or malfunction.

The permittee shall operate the engine gen-sets and SCR such that the exhaust temperature at the outlet of the entire SCR+cDPF emissions control system does not exceed 1,022°F. The SCR shall be provided with adequate access for inspection and shall be in operation when the engine gen-sets are operating as stated above.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

3. **Emission Controls (cDPF)** – Particulate matter (PM, PM<sub>10</sub>, and PM<sub>2.5</sub>), carbon monoxide (CO), and volatile organic compounds (VOC) emissions from the non-emergency diesel engine gen-sets (Ref. Nos. 1 through 22) shall be controlled by catalyzed Diesel Particulate Filter (cDPF). The cDPF shall be provided with adequate access for inspection and shall be in operation when the non-emergency diesel engine gen-sets are operating.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

4. **Monitoring Devices (SCR & cDPF)** – Each non-emergency diesel engine gen-set (Ref. No. 1 through 22) shall be equipped with devices to continuously measure and record the following, as applicable:

- a. The exhaust temperature at the outlet of the entire SCR+cDPF emissions control system. The information shall be recorded at a minimum frequency of once every

fifteen minutes and correlated to run date, engine operating hours, and fuel consumption.

- b. The NO<sub>x</sub> concentration measured after the SCR catalyst, expressed in ppm. The information shall be recorded at a minimum frequency of once every fifteen minutes and correlated to run date, engine operating hours, and fuel consumption.
- c. The differential pressure drop across the entire SCR+cDPF emissions control system to demonstrate proper function. The backpressure monitor shall notify the permittee when the high backpressure limit of the engine is approached. The differential pressure drop shall be recorded by the permittee with a frequency as recommended by the process/control equipment manufacturer.
- d. The exhaust temperature at the inlet of the entire SCR+cDPF emissions control system. The information shall be recorded at a minimum frequency of once every fifteen minutes and correlated to run date, engine operating hours, and fuel consumption.

Each monitoring device shall be installed, maintained, calibrated and operated in accordance with approved procedures which shall include, at a minimum, the manufacturer's written requirements or recommendations. Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the SCR and cDPF are operating.

(9VAC5-80-1180 D) [3/26/2026]

## **5. Monitoring –**

- a. Fuel Flow: Each diesel engine gen-set (Ref. Nos. 1 through 23, T1 and T2) shall be equipped with a device to continuously measure and record individual fuel consumption (in gallons) for each diesel engine gen-set.
- b. Engine Operating Hours: Each diesel engine gen-set (Ref. Nos. 1 through 23, T1 and T2) shall be equipped with a non-resettable hour meter which measures the duration of time that each engine gen-set is operated.

Each monitoring device (as required in a. through b. above) shall be observed by the permittee with a frequency of not less than once each day the diesel engine gen-set is operated. The permittee shall keep a log of these observations.

Each monitoring device shall be installed, maintained, calibrated (as appropriate), and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. The details of the monitoring device calibrations are to be arranged with the Regional Air Compliance Manager of the DEQ's Northern Regional Office (NRO).

Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the engine gen-sets are operating.  
(9VAC5-80-1180 D) [3/26/2026]

## **OPERATING/EMISSION LIMITATIONS**

- 6. Emergency Power Generation** – The emergency diesel engine gen-sets (Ref. Nos. 23, T1 and T2) shall only be operated in the following modes:
- a. In situations that arise from sudden and reasonably unforeseeable events where the primary energy or power source is disrupted or disconnected due to conditions beyond the control of an owner or operator of a facility including:
    - i. A failure of the electrical grid;
    - ii. On-site disaster or equipment failure; or
    - iii. Public service emergencies such as flood, fire, natural disaster, or severe weather conditions.
  - b. For participation in an independent system operator (ISO)-declared emergency, where an ISO emergency is:
    - i. An abnormal system condition requiring manual or automatic action to maintain system frequency, to prevent loss of firm load, equipment damage, or tripping of system elements that could adversely affect the reliability of an electric system or the safety of persons or property;
    - ii. Capacity deficiency or capacity excess conditions;
    - iii. A fuel shortage requiring departure from normal operating procedures in order to minimize the use of such scarce fuel;
    - iv. Abnormal natural events or man-made threats that would require conservative operations to posture the system in a more reliable state; or
    - v. An abnormal event external to the ISO service territory that may require ISO action.
  - c. For scheduled maintenance checks and readiness testing (Scheduled MCRT).
  - d. For unscheduled maintenance, testing, and operational training.

- e. For the integration operational period, which is the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source's electrical system.

(9VAC5-80-1180) [3/26/2026]

- 7. **Operation of the Engine Gen-Sets** – The permittee shall operate and maintain the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) and control device according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not increase air emissions.  
(9VAC5-80-1180) [3/26/2026]
- 8. **Operating Limitations (Ozone Season)** – No diesel engine gen-set (Ref. Nos. 1 through 23, T1 and T2) shall be operated for scheduled maintenance checks and readiness testing (Scheduled MCRT), stack testing, or operational training (that involves fuel combustion) between the hours of 7 a.m. to 5 p.m. any day during May 1 through September 30. The permittee may petition the Regional Air Compliance Manager of DEQ's NRO, for exceptions to this requirement, with approvals made on a case-by-case basis.  
(9VAC5-80-1180) [3/26/2026]
- 9. **Operating Limitations (Ozone Season) – Integration Operational Period** – During the integration operational period of the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2), any operation of the unit (that involves fuel combustion) between the hours of 7 a.m. to 5 p.m. any day during the ozone season of May 1 through September 30 shall only occur if the forecast Air Quality Index (AQI) for ozone as published on the AirNow website (<https://airnow.gov>) for Northern Virginia for that day is less than or equal to 100. In the event that AirNow-EnviroFlash ([www.enviroflash.info](http://www.enviroflash.info)) issues an Air Alert for Metropolitan Washington, D.C. for a day which the forecasted AQI for ozone was less than or equal to 100, operation of each unit (which involves fuel combustion) shall be minimized to the maximum extent practical.  
(9VAC5-80-1180) [3/26/2026]
- 10. **Operating Hours** – The operating hours of the emergency diesel engine gen-sets are limited to the following:
  - a. No single unit (Ref. Nos. 23, T1 and T2) shall operate more than 500 hours per year for all purposes (as provided in Condition 6), calculated monthly as the sum of each consecutive 12-month period.
  - b. No single unit (Ref. Nos. 23, T1 and T2) shall operate more than 35 hours per year for scheduled maintenance checks and readiness testing (Scheduled MCRT, as provided in Condition 6.c).

Compliance for each consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.  
(9VAC5-80-1180) [3/26/2026]

- 11. Fuel Specification** – The approved fuel for the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) is diesel fuel oil. For the purposes of this permit document, diesel fuel oil is defined as ultra-low sulfur diesel fuel oil (ULSD), renewable diesel, or a blend of these fuels, and shall meet the specifications below:

**DIESEL FUEL OIL:**

- a. Does not exceed the American Society for Testing and Materials (ASTM) specification, D975, for grade ultra-low sulfur 1-D S15 or grade 2-D S15, or
- b. Has a maximum sulfur content not to exceed 0.0015% by weight (15 ppm), and either a minimum cetane number of 40 or maximum aromatic content of 35 volume percent.

Exceedance of these specifications may be considered credible evidence of an exceedance of emission limits. A change in the fuel type or the fuel sulfur content may require a permit to modify and operate.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

- 12. Fuel Certification** – The permittee shall obtain a certification from the fuel supplier with each shipment of diesel fuel oil. Each fuel supplier certification shall include the following:

- a. The name of the fuel supplier;
- b. The date on which the diesel fuel oil was received;
- c. The quantity of diesel fuel oil delivered in the shipment; and
- d. A statement that the diesel fuel oil complies with the requirements of Condition 11 Fuel Specification.

Alternatively, the permittee must obtain approval from the Regional Air Compliance Manager of the DEQ's NRO, if other documentation will be used to certify the diesel fuel oil type.

Fuel sampling and analysis, independent of that used for certification, as may be periodically required or conducted by DEQ, may be used to determine compliance with the fuel specifications stipulated in Condition 11.

(9VAC5-80-1180) [3/26/2026]

- 13. Fuel Throughput Limit** – The diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) for all operations shall only consume a combined quantity of diesel fuel oil (in gallons) each consecutive 365-day period, as demonstrated by using the following equation:

$$\frac{\text{Group 1 Fuel Consumption}}{8,360,462 \text{ gallons}} + \frac{\text{Group 2 Fuel Consumption}}{696,705 \text{ gallons}} + \frac{\text{Group 3 Fuel Consumption}}{717,805 \text{ gallons}} + \frac{\text{Group 4 Fuel Consumption}}{682,223 \text{ gallons}} + \leq 1$$

Where:

| Engine Group | Ref. Nos.    | Engine Model<br>(SCR & cDPF Status)                       |
|--------------|--------------|-----------------------------------------------------------|
| Group 1      | 1 through 22 | Caterpillar 3516E<br>(SCR & cDPF Controlled)              |
| Group 2      | 1 through 22 | Caterpillar 3516E<br>(SCR Uncontrolled & cDPF Controlled) |
| Group 3      | 23           | Caterpillar C18<br>(No Controls)                          |
| Group 4      | T1 and T2    | Caterpillar 3516C<br>(No Controls)                        |

Compliance for the consecutive 365-day period shall be demonstrated daily by adding the total for the most recently completed calendar day to the individual daily totals for the preceding 364 days.

(9VAC5-80-1180) [3/26/2026]

## EMISSION LIMITS

- 14. Emission Limits (Hourly – Emergency)** – Emissions from the operation of each emergency diesel engine gen-set (Ref. Nos. 23, T1 and T2) shall not exceed the limits specified below:

| <b>Pollutant</b>                                         | <b>750 ekW Caterpillar<br/>C18<br/>Group 3<br/>(Ref. No. 23)</b> | <b>2,000 ekW Caterpillar<br/>3516C<br/>Group 4<br/>(Ref. Nos. T1 and T2)</b> |
|----------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------|
| Nitrogen Oxides<br>(NO <sub>x</sub> as NO <sub>2</sub> ) | 14.34 lbs/hr                                                     | 38.85 lbs/hr                                                                 |
| Carbon Monoxide (CO)                                     | 1.28 lbs/hr                                                      | 3.92 lbs/hr                                                                  |
| Volatile Organic Compounds<br>(VOC)                      | 0.28 lbs/hr                                                      | 1.14 lbs/hr                                                                  |
| Particulate Matter (PM)                                  | 0.15 lbs/hr                                                      | 0.57 lbs/hr                                                                  |
| Particulate Matter (PM <sub>10</sub> )                   | 0.15 lbs/hr                                                      | 0.57 lbs/hr                                                                  |
| Particulate Matter (PM <sub>2.5</sub> )                  | 0.15 lbs/hr                                                      | 0.57 lbs/hr                                                                  |
| Sulfur Dioxide (SO <sub>2</sub> )                        | 0.01 lbs/hr                                                      | 0.03 lbs/hr                                                                  |

Compliance with these emission limits shall be based on the proper operation and maintenance of the emergency diesel engine gen-sets or by testing, if required.  
(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

- 15. Emission Limits (Hourly – Non-Emergency)** – Emissions from the operation of each non-emergency diesel engine gen-sets (Ref. Nos. 1 through 22) shall not exceed the limits specified below:

| Pollutant                                             | 2,750 kW Caterpillar 3516E<br>(Groups 1 and 2)<br>(Ref. Nos. 1 through 22) |              |
|-------------------------------------------------------|----------------------------------------------------------------------------|--------------|
|                                                       | Controlled                                                                 | Uncontrolled |
| Nitrogen Oxides (NO <sub>x</sub> as NO <sub>2</sub> ) | 4.46 lbs/hr*                                                               | 53.48 lbs/hr |
| Carbon Monoxide (CO)                                  | 2.05 lbs/hr*                                                               |              |
| Volatile Organic Compounds (VOC)                      | 0.45 lbs/hr*                                                               |              |
| Particulate Matter (PM)                               | 0.18 lbs/hr*                                                               |              |
| Particulate Matter (PM <sub>10</sub> )                | 0.18 lbs/hr*                                                               |              |
| Particulate Matter (PM <sub>2.5</sub> )               | 0.18 lbs/hr*                                                               |              |
| Sulfur Dioxide (SO <sub>2</sub> )                     | 0.04 lbs/hr                                                                |              |

\*Exhaust gas treated with SCR+cDPF emissions control

Compliance with these emissions limits shall be based on the proper operation and maintenance of the non-emergency diesel engine gen-sets, pollution control devices, or by testing, if required.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

- 16. Emission Limits (Annual)**– Emissions from the operation of the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) shall not exceed the limits specified below:

| Pollutant                                             | Groups 1 through 4<br>(Ref. Nos. 1 through 23, T1 and T2) |
|-------------------------------------------------------|-----------------------------------------------------------|
| Nitrogen Oxides (NO <sub>x</sub> as NO <sub>2</sub> ) | 96.03 tpy                                                 |
| Carbon Monoxide (CO)                                  | 55.99 tpy                                                 |
| Volatile Organic Compounds (VOC)                      | 24.70 tpy                                                 |
| Particulate Matter (PM)                               | 3.95 tpy                                                  |
| Particulate Matter (PM <sub>10</sub> )                | 3.95 tpy                                                  |
| Particulate Matter (PM <sub>2.5</sub> )               | 3.95 tpy                                                  |
| Sulfur Dioxide (SO <sub>2</sub> )                     | 0.87 tpy                                                  |

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with the annual emission limits may be determined as stated in Conditions 1, 2, 3, 4, 13, 14 and 15.

(9VAC5-80-1180) [3/26/2026]

#### **17. Visible Emission Limit –**

- a. Visible emissions from each non-emergency engine gen-set (Ref. Nos. 1 through 22) shall not exceed 5% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A). This requirement applies at all times except during startup, shutdown and malfunction.
- b. Visible emissions from each emergency engine gen-set (Ref. Nos. 23, T1 and T2) shall not exceed 5% opacity except during one six-minute period in any one hour in which visible emissions shall not exceed 10% opacity as determined by the EPA Method 9 (reference 40 CFR 60, Appendix A). This requirement applies at all times except during startup, shutdown and malfunction.

During startup and shutdown, visible emissions from each engine gen-set shall not exceed 10% opacity except during one (1) six-minute period in any one-hour in which visible emissions shall not exceed 20% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

#### **INITIAL COMPLIANCE DETERMINATION**

- 18. Stack Tests –** Initial performance tests shall be conducted on four (4) of the Caterpillar 3516E non-emergency diesel engine gen-sets (Ref. Nos. 1 through 22) for NO<sub>x</sub> (as NO<sub>2</sub>) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the controlled emission limits contained in Condition 15.

- a. Emissions testing of each pollutant for each selected non-emergency diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3) runs shall be reported as the short-term emission rate for that non-emergency diesel engine gen-set;
- b. Testing shall be performed on the exhaust stack of the non-emergency diesel engine gen-sets to demonstrate compliance with the controlled NO<sub>x</sub> and CO emission limits specified in Condition 15. Testing shall be conducted with the non-emergency diesel engine gen-set operating at  $\geq 90$  percent of its rated capacity, unless multiple load band testing is approved by DEQ;

- c. Recorded non-emergency engine gen-set operational information shall include, but not be limited to:
  - i. Generator load/kilowatt output;
  - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil;
  - iii. NO<sub>x</sub> concentration after the SCR + cDPF emissions control system;
  - iv. Exhaust temperature at the outlet of the entire SCR + cDPF emissions control system;
  - v. Urea solution injection rate; and
  - vi. Differential pressure drop across the entire SCR + cDPF emissions control system.
- d. Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. The integration operational period is defined as: the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source electrical system. If this deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- e. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- f. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- g. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [3/26/2026]

- 19. Stack Tests (Renewable Diesel)** – Performance tests shall be conducted on two (2) diesel engine gen-sets (Ref. Nos. 1 through 22) while utilizing renewable diesel or a blend of renewable diesel and ULSD for NO<sub>x</sub> (as NO<sub>2</sub>) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the controlled emission limits contained in Condition 15.
- a. Emissions testing of each pollutant for each selected diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3) runs shall be reported as the short-term emission rate for that diesel engine gen-set.
  - b. Testing shall be performed on the exhaust stack of the diesel engine gen-sets to demonstrate compliance with the controlled NO<sub>x</sub> and CO emission limits specified in Condition 15. Testing shall be conducted with the diesel engine gen-set operating at ≥ 90 percent of its rated capacity, unless multiple load band testing is approved by DEQ;
  - c. Recorded diesel engine gen-set operational information shall include, but not be limited to:
    - i. Generator load/kilowatt output;
    - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil;
  - d. For any engine gen-set that has not commenced the integration operational period upon receipt of the first shipment of renewable diesel or a blend of renewable diesel and ULSD utilized by the units: Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. If this deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
  - e. For any engine gen-set that has commenced the integration operational period upon receipt of the first shipment of renewable diesel or a blend of renewable diesel and ULSD utilized by the units: Perform testing to demonstrate compliance within 120 days of the renewable diesel fuel or a blend of renewable diesel and ULSD first utilized by the affected units. The permittee may petition the DEQ's NRO Air Compliance Manager for an extension to this deadline, with approvals made on a case by-case basis. If the applicable deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
  - f. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less

than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;

- g. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- h. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [3/26/2026]

- 20. Visible Emissions Evaluation** – Concurrent with the initial performance tests required in Conditions 18 and 19, Visible Emission Evaluations (VEE) in accordance with 40 CFR Part 60, Appendix A, Method 9, shall also be conducted by the permittee on the diesel engine gen-sets selected for initial performance testing. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit a VEE protocol in conjunction with the initial stack test protocol required by Conditions 18 and 19, at least 30 days prior to testing.

- a. Should conditions prevent concurrent opacity observations, the Regional Air Compliance Manager of the DEQ's NRO shall be notified in writing, within seven (7) days, and visible emissions testing shall be rescheduled within 30 days. Rescheduled testing shall be conducted under the same conditions (as possible) as the initial performance tests.
- b. Two (2) copies of the test result (one (1) hard copy and one (1) electronic copy) shall be submitted to the Regional Air Compliance Manager of the DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit (Attachment A).

(9VAC5-50-30 and 9VAC5-80-1200) [3/26/2026]

## **CONTINUING COMPLIANCE DETERMINATION**

- 21. Facility Construction** – The diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) shall be constructed so as to allow for emissions testing upon reasonable notice at any times, using appropriate methods. This includes constructing the facility/equipment such that volumetric flow rates and pollutant emission rates can be determined by applicable test methods and providing a stack or duct that is free from cyclonic flow. Sampling ports shall be provided when requested at the appropriate locations and safe sampling platforms and access shall be provided.

(9VAC5-50-30 F and 9VAC5-80-1180) [3/26/2026]

- 22. Emission Testing/Visible Emissions Evaluation** – Upon request by the DEQ, the permittee shall conduct stack tests and/or VEEs on the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) to demonstrate compliance with the emission limits contained in this permit. The details of the tests shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO.  
(9VAC5-80-1200 and 9VAC5-50-30 G) [3/26/2026]

## RECORDS

- 23. On Site Records** – The permittee shall maintain records of emission data and operating parameters as necessary to demonstrate compliance with this permit. The content and format of such records shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO. These records shall include, but are not limited to:
- a. Operation and control system monitoring records for each non-emergency diesel engine gen-set (Ref. Nos. 1 through 22) equipped with an SCR as required in Condition 4.a. and 4.b. This includes records of the exhaust temperature at the outlet of the entire SCR + cDPF emissions control system and NO<sub>x</sub> concentration as measured by the SCR continuous monitoring device.
  - b. Operation and control device monitoring records for each non-emergency engine gen-set (Ref. Nos. 1 through 22) equipped with a cDPF as required in Condition 4.c. This includes records of the differential pressure drop across the entire SCR + cDPF emissions control system.
  - c. Operation and control device monitoring records for each non-emergency engine gen-set (Ref. Nos. 1 through 22) equipped with a cDPF as required in Condition 4.d. This includes records of the exhaust temperature at the inlet of the entire SCR + cDPF emissions control system.
  - d. A log of monitoring device observations as required by Condition 5.
  - e. Records of the reasons for operation for each emergency diesel engine gen-set (Ref. Nos. 23, T1 and T2), including, but not limited to, the date, cause of operation, cause of the emergency, the ISO-declared emergency notification, and the hours of operation.
  - f. Records for diesel engine gen-set operations, as necessary, to demonstrate compliance with the operating limitations of Condition 8; which includes but is not limited to: times, dates and reasons for operation of the diesel engine gen-set that was operating between May 1 and September 30.
  - g. To verify compliance with Condition 9, maintain records of:

- i. The forecasted AQI, as determined by the AirNow website for Northern Virginia, for ozone for the day(s) that a diesel engine gen-set operated during the integration operational period;
  - ii. The measured AQI, as determined by the AirNow website for Northern Virginia, for ozone for the day(s) that the diesel engine gen-set operated during the integration operational period;
  - iii. Documentation recording any Air Alerts issued for that operating day, as determined by AirNow-EnviroFlash; and
  - iv. Details of Integration Operational Period (IOP) activities, to include, but not limited to, clock hours, and duration.
- h. Monthly and annual hours of operation of each emergency diesel engine gen-set (Ref. Nos. 23, T1 and T2), calculated monthly as the sum of each consecutive rolling 12-month period, as required in Condition 10.a.
- i. Monthly and annual hours of operation of each emergency diesel engine gen-set (Ref. Nos. 23, T1 and T2), for purposes of Scheduled MCRT, with annual hours of operation calculated monthly as the sum of each consecutive 12-month period, as required in Condition 10.b.
- j. All fuel supplier certifications, as required in Condition 12.
- k. Daily and annual fuel consumption of each diesel engine gen-set (Ref. No. 1 through 23, T1 and T2), calculated daily as the sum of each consecutive 365-day period, as required in Condition 13.
- l. Daily and annual fuel consumption for each diesel engine gen-set by group (Groups 1 through Group 4), calculated daily as the sum of each consecutive 365-day period for the purposes of the compliance demonstration with the diesel fuel throughput limit equation in Condition 13.
- m. Daily and annual results of the computed fuel throughput equation used by the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) by groups (Group 1 through Group 4), calculated daily as the sum of each consecutive 365-day period for the purposes of the compliance demonstration with the diesel fuel throughput limit equation in Condition 13.
- n. Daily and annual emissions calculations for NO<sub>x</sub> (as NO<sub>2</sub>), CO, VOC, PM, PM<sub>10</sub>, PM<sub>2.5</sub>, and SO<sub>2</sub> from the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2) for all operations, combined, to verify compliance with the annual emission limitations

in Condition 16, with annual emissions calculated daily as the sum of each consecutive 365-day period.

- o. Results of all stack tests and VEEs.
- p. Records of scheduled and unscheduled maintenance, operator training, and records as required by Condition 31.
- q. Documentation from the manufacturer that the diesel engine gen-sets are certified to meet the EPA Tier 2 emission standards.
- r. Engine information including make, model, serial number, model year, maximum engine power (bhp), and engine displacement for the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2).
- s. Records of the manufacturer's written instructions or procedures developed by the owner or operator that are approved by the engine manufacturer.
- t. Records of changes in settings that are permitted by the manufacturer of the diesel engine gen-sets (Ref. Nos. 1 through 23, T1 and T2).

Compliance for the consecutive 12-month period in the subsections above (as applicable) shall be demonstrated monthly by adding the total for the most recently completed month to the individual monthly totals for the preceding 11 months.

Compliance for the consecutive 365-day period in the subsections above (as applicable) shall be demonstrated daily by adding the total for the most recently completed day to the individual daily totals for the preceding 364 days.

These records shall be available for inspection by the DEQ and shall be current for the most recent five (5) years, unless otherwise noted  
(9VAC5-80-1180 and 9VAC5-50-50) [3/26/2026]

## **SPECIAL CONDITIONS – TRANSITORY ENGINE-GENERATOR SETS**

- 24. Operation of the Transitory Engine Gen-Sets** – The facility shall only operate the transitory emergency diesel engine gen-sets (Ref. Nos. T1 and T2) in support of the facility such as serving as back up during construction, commissioning, and maintenance of the engine gen-sets (Ref. Nos. 1 through 23).  
(9VAC5-80-1180) [3/26/2026]
- 25. Notifications** – The permittee shall furnish the following written notifications to the Regional Air Compliance Manager of the DEQ's NRO of:

- a. The actual date and reason for each occurrence that each transitory emergency diesel engine gen-set (Ref. Nos. T1 and T2) was placed into service within 15 days after such date. The notification must include the following:
  - i. Name and address of the permittee;
  - ii. The address of the affected source;
  - iii. Engine information including make, model, engine family, serial number, model year, maximum engine power and engine displacement;
  - iv. Fuel used; and
  - v. Hours operated.
- b. The actual date of the removal of each transitory emergency diesel engine gen-set (Ref. Nos. T1 and T2) within 15 days after such date.

(9VAC5-80-1180) [3/26/2026]

## NOTIFICATIONS

- 26. Initial Notifications** – The permittee shall furnish written notification of the items below to the Air Compliance Manager of the DEQ’s Northern Regional Office at the following address:

Regional Air Compliance Manager  
Department of Environmental Quality  
13901 Crown Court  
Woodbridge, VA 22193

The permittee shall submit one notification for each building containing the information as described below:

- a. The actual date on which installation of the engine gen-sets (Ref. Nos. 1 through 23) commenced in the building, within thirty (30) days after such date. The notification must contain the following:
  - i. Name and address of the permittee;
  - ii. The building;
  - iii. Unit reference number of the initial unit installed; and
  - iv. The date installation commenced.

- b. The date that the integration operational period started for each diesel engine gen-set (Ref. Nos. 1 through 23) within fifteen (15) days after the last generator at each building, or construction phase, completes its integration operational period. If a period of construction is paused or halted for  $\geq 45$  days, this notification shall be provided to the DEQ within fifteen (15) days after completion of the integration operational period for the most recently installed diesel engine gen-set. The notification must contain the following:
  - i. Unit reference number and engine group;
  - ii. Engine information including make, model, engine family, serial number, model year, maximum engine power, engine displacement, fuel used;
  - iii. Installation date; and
  - iv. Integration operational period start and end dates.

For the purpose of this notification, the integration operational period is defined as: the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source's electrical system.  
(9VAC5-50-20) [3/26/2026]

**27. Renewable Diesel Notification** – Upon receipt of the first shipment of renewable diesel or a blend of renewable diesel and ULSD, the permittee shall furnish written notification of the items below to the Air Compliance Manager of the DEQ's NRO.

- a. The actual date on which the shipment was received within fifteen (15) days after such date. The notification must include the following:
  - i. Name and address of the permittee;
  - ii. The address of the affected source;
  - iii. Engine gen-sets (with reference numbers) utilizing the fuel in the shipment; and
  - iv. Fuel certification (as provided in Condition 12)

(9VAC5-80-1180) [3/26/2026]

**GENERAL CONDITIONS**

**28. Permit Invalidation** – This permit to construct the diesel engine gen-sets (Ref. Nos. 1 through 23) shall become invalid, unless an extension is granted by the DEQ, if:

- a. A program of continuous construction is not commenced within 18 months from 'Original Permit Date' specified in the equipment list in the Introduction section of this permit; or if
  - b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time, except for a DEQ approved period between phases of the phased construction of a new stationary source or project.
- (9 VAC 5-80-1210)

**29. Permit Suspension/Revocation** – This Board may suspend or revoke any permit if the permittee:

- a. Knowingly makes material misstatements in the permit application or any amendments to it;
- b. Fails to comply with the conditions of this permit;
- c. Fails to comply with any emission standards applicable to a permitted emissions unit;
- d. Causes emissions from the stationary source which result in violations of, or interfere with the attainment and maintenance of, any ambient air quality standard; or
- e. Fails to operate in conformance with any applicable control strategy, including any emission standards or emissions limitations, in the State Implementation Plan in effect at the time an application for this permit is submitted.

(9VAC5-80-1210 G)

**30. Right of Entry** – The permittee shall allow authorized local, state, and federal representatives, upon the presentation of credentials:

- a. To enter upon the permittee's premises on which the facility is located or in which any records are required to be kept under the terms and conditions of this permit;
- b. To have access to and copy at reasonable times any records required to be kept under the terms and conditions of this permit or the State Air Pollution Control Board Regulations;
- c. To inspect at reasonable times any facility, equipment, or process subject to the terms and conditions of this permit or the State Air Pollution Control Board Regulations; and
- d. To sample or test at reasonable times.

For purposes of this condition, the time for inspection shall be deemed reasonable during regular business hours or whenever the facility is in operation. Nothing contained herein shall make an inspection time unreasonable during an emergency.  
(9VAC5-170-130 and 9VAC5-80-1180)

- 31. Maintenance/Operating Procedures** – At all times, including periods of start-up, shutdown, and malfunction, the permittee shall, to the extent practicable, maintain and operate the affected source, including associated air pollution control equipment, in a manner consistent with good air pollution control practices for minimizing emissions.

The permittee shall take the following measures in order to minimize the duration and frequency of excess emissions:

- a. Develop a maintenance schedule and maintain records of all scheduled and non-scheduled maintenance.
- b. Maintain an inventory of spare parts.
- c. Have available written operating procedures for equipment. These procedures shall be based on the manufacturer's recommendations, at a minimum.
- d. Train operators in the proper operation of all such equipment and familiarize the operators with the written operating procedures, prior to their first operation of such equipment. The permittee shall maintain records of the training provided including the names of trainees, the date of training and the nature of the training.

Records of maintenance and training shall be maintained on site for a period of five years and shall be made available to DEQ personnel upon request.  
(9VAC5-50-20 E and 9VAC5-80-1180 D)

- 32. Record of Malfunctions** – The permittee shall maintain records of the occurrence and duration of any bypass, malfunction, shutdown or failure of the facility or its associated air pollution control equipment that results in excess emissions for more than one hour. Records shall include the date, time, duration, description (emission unit, pollutant affected, cause), corrective action, preventive measures taken and name of person generating the record.  
(9VAC5-20-180 J and 9VAC5-80-1180 D)

- 33. Notification for Facility or Control Equipment Malfunction** – The permittee shall furnish notification to the Regional Air Compliance Manager of the DEQ's NRO of malfunctions of the affected facility or related air pollution control equipment that may cause excess emissions for more than one hour. Such notification shall be made no later than four daytime business hours after the malfunction is discovered. The permittee shall provide a written statement giving all pertinent facts, including the estimated duration of the breakdown, within 14 days of discovery of the malfunction. When the condition causing

the failure or malfunction has been corrected and the equipment is again in operation, the permittee shall notify the Regional Air Compliance Manager of the DEQ's NRO.  
(9VAC5-20-180 C and 9VAC5-80-1180)

- 34. Violation of Ambient Air Quality Standard** – The permittee shall, upon request of the DEQ, reduce the level of operation or shut down a facility, as necessary to avoid violating any primary ambient air quality standard and shall not return to normal operation until such time as the ambient air quality standard will not be violated.  
(9VAC5-20-180 I and 9VAC5-80-1180)
- 35. Change of Ownership** – In the case of a transfer of ownership of a stationary source, the new owner shall abide by any current minor NSR permit issued to the previous owner. The new owner shall notify the NRO of the change of ownership within 30 days of the transfer.  
(9VAC5-80-1240)
- 36. Permit Copy** – The permittee shall keep a copy of this permit on the premises of the facility to which it applies.  
(9VAC5-80-1180)

## APPENDIX A: SOURCE TESTING REPORT FORMAT

### Report Cover

1. Plant name and location
2. Units tested at source (indicate Ref. No. used by source in permit or registration)
3. Test Dates.
4. Tester; name, address and report date

### Certification

1. Signed by team leader/certified observer (include certification date)
2. Signed by responsible company official
3. \*Signed by reviewer

### Copy of approved test protocol

### Summary

1. Reason for testing
2. Test dates
3. Identification of unit tested & the maximum rated capacity
4. \*For each emission unit, a table showing:
  - a. Operating rate
  - b. Test Methods
  - c. Pollutants tested
  - d. Test results for each run and the run average
  - e. Pollutant standard or limit
5. Summarized process and control equipment data for each run and the average, as required by the test protocol
6. A statement that test was conducted in accordance with the test protocol or identification & discussion of deviations, including the likely impact on results
7. Any other important information

### Source Operation

1. Description of process and control devices
2. Process and control equipment flow diagram
3. Sampling port location and dimensioned cross section Attached protocol includes: sketch of stack (elevation view) showing sampling port locations, upstream and downstream flow disturbances and their distances from ports; and a sketch of stack (plan view) showing sampling ports, ducts entering the stack and stack diameter or dimensions

### Test Results

1. Detailed test results for each run
2. \*Sample calculations
3. \*Description of collected samples, to include audits when applicable

### Appendix

1. \*Raw production data
2. \*Raw field data
3. \*Laboratory reports
4. \*Chain of custody records for lab samples
5. \*Calibration procedures and results
6. Project participants and titles
7. Observers' names (industry and agency)
8. Related correspondence
9. Standard procedures

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\* Not applicable to visible emission evaluations